CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 3 : Python Program for Water Jug Problem

Aim

To develop a Python program that measures a required amount of water using two jugs of given capacities.

Algorithm

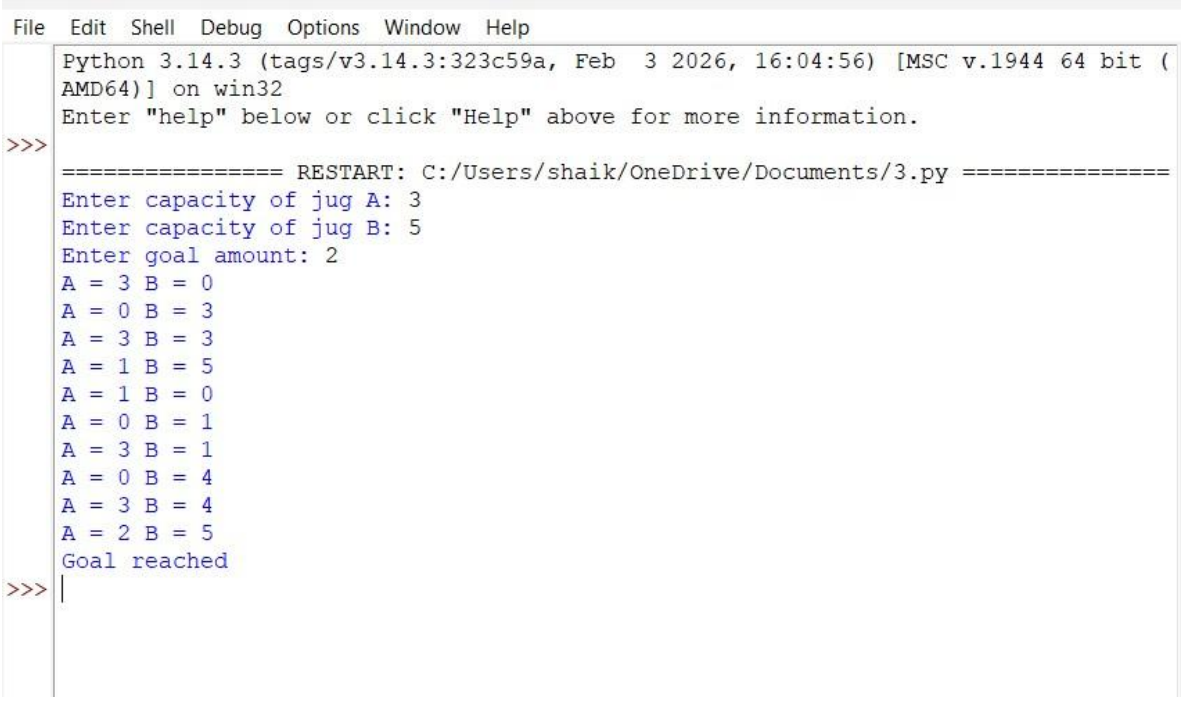
- Read the capacity of jug A, the capacity of jug B and the goal amount.
- Start with both jugs empty (A = 0, B = 0).
- Repeat until either jug contains the goal amount.
- If jug A is empty, fill jug A completely.
- Otherwise, if jug B is full, empty jug B.
- Otherwise, pour water from jug A into jug B until jug A is empty or jug B is full.
- Print the amount of water in both jugs after every step.
- When the goal amount is reached, print "Goal reached".

Program

```
a = int(input("Enter capacity of jug A: "))
b = int(input("Enter capacity of jug B: "))
goal = int(input("Enter goal amount: ")) x
= 0 y = 0

while x != goal and y != goal:
if x == 0:          x = a
elif y == b:          y = 0
else:
        t = min(x, b -
y)          x = x - t
y = y + t     print("A =", x,
"B =", y) print("Goal
reached")
```

Output



RESULT

Thus, the Water Jug Problem was solved successfully using Python.